

SHASHWAT DUDEJA | Software Engineer

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EDUCATION

University of California, San Diego — B.S. Mathematics–Computer Science

Sep 2022 – May 2026

Relevant Coursework: Advanced Data Structures & Algorithms, Machine Learning, AI Agents, Computer Organization, Software Engineering, Front-End Web Development, Back-End Web Development.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, HTML5, CSS, SQL

Frameworks & Tools: React, React-native, PyTorch, Django, FastAPI, NumPy, Pandas, Git, Github

WORK EXPERIENCE

Software Engineering Intern — Mashreq Bank, Dubai, UAE

Jul 2023 – Aug 2023

- Developed an algorithm to parse JSON-based financial data for customer KYC verification, ensuring clean and scalable code for high-volume processing.
- Proposed UI/UX and infrastructure improvements for smoother client operations, researching and designing enhancements to existing software systems.
- Trained new interns in company workflows and onboarded them, contributing to technical documentation and staff training protocols.

Machine Learning Intern — Mercury Alert AI, San Diego

Nov 2025 – Jul 2026

- Benchmarked and fine-tuned computer vision models with synthetic data for live camera fall recognition to increase real-time fall detection accuracy on live camera feeds.
- Implemented 2FA and Google authentication on mobile applications and web portals to enhance security systems and user data protection.
- Implemented automated email feature for caregivers for safety score anomalies and potential emergencies, evaluating user feedback to recommend program improvements.

PROJECTS

[Chess AI](#) | C++, WebAssembly, React.js, CSS

- Developed a C++ chess engine (compiled to WASM) with a react UI performing at 2000 Elo.
- Implemented alpha-beta pruning, transposition tables, bitboard representations, magic bitboards(for O(1) move generation).

[Full-Stack Web Analytics & Reporting Engine](#) | Node.js, Express, MySQL, Chart.js, HTML5, JavaScript, CSS

- Developed a full-stack web analytics platform featuring multi-tier Role-Based Access Control (RBAC).
- Engineered relational MySQL database schemas and Express routes to log user activity, generating dynamic data visualisations using Chart.js and automated PDF export reports.
- Engineered an automated CI/CD pipeline using GitHub Actions to continuously deploy updates via SSH to a multi-vhost Apache server hosted on an Ubuntu instance.

[NBA Team Performance Modeling with Hidden Markov Models](#) | Python, NumPy, Pandas, Matplotlib

- Developed a probabilistic agent that models sequential evolution of NBA team performance using HMMs to predict game outcomes.
- Implemented Baum-Welch training and Viterbi decoding in Python to infer hidden team performance states and predict win probabilities.

[Connect4 AI](#) | HTML5, JavaScript, CSS, C++, WebAssembly

- Built a benchmarking site with several different algorithms and implementations to note amount of improvement and significance from each feature and optimization.
- Engineered bitboard state representations, heuristic evaluations, different search functions to optimize connect 4 solver, benchmarking node evaluations across algorithmic variations.

[Generative Music & Audio Synthesis Model](#) | Python, PyTorch, Librosa, NumPy, SciPy, Pandas

- Built an unconditioned note language model using a pitch-token Gated Recurrent Unit (GRU) in PyTorch to generate symbolic MIDI music sequences.
- Processed a dataset of ~1,599 classical piano excerpts; quantized note timing into 120-tick steps and converted compositions into pitch-token sequences across a 128-pitch vocabulary (including BOS/EOS/PAD tokens).